



HACKWELL 2.0



FARMCHAIN AI: TRANSPARENT FARM-TO-CONSUMER MARKETPLACE

TEAM NAME

The Knowledge Knights

TEAM ID

H20-165

PROBLEM STATEMENT NUMBER(PS ID)

CDT-09

DOMAIN

Precision Farming · Cybersecurity & Digital Trust

TEAM MEMBERS

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Problem Statement

Blockchain proves a record is unchanged, not that it is true

◆ Trust gaps in the farm-to-consumer chain

Farmers lose value to opaque intermediaries, and even blockchain tracing fails on three gaps: fake crop data (GIGO), photocopied QR codes, and wallet / gas-fee friction.

◆ Who is affected?

Farmers face unfair prices and delayed payments. Consumers and retailers cannot verify the origin or quality of “premium” or “organic” produce.

◆ Current solutions & limitations

Paper certificates can be forged. Typical blockchain trackers record unverified inputs, allow copied QR codes, and expect farmers to manage crypto wallets.

Problem Impact & Scale

- **Most affected:** small farmers in India, and buyers of premium / organic produce
- **Key data point:** 40–60% of crop value lost to markups, fraud and payment delays*
- **Frequency:** every harvest and every sale; each cloned QR scan repeats the fraud

**Figure from our project brief; add source before final submission.*

Proposed Solution

One platform that verifies the crop, the QR and the payment

◆ What it is and how it works

FarmChain AI is a decentralised farm-to-consumer marketplace. AI scores each crop, smart contracts record it on-chain, and escrow pays the farmer automatically on delivery.

◆ How it addresses the problem

An in-browser AI Visual Oracle blocks fake crop data, spatial-temporal QR checks detect cloned labels, and phone-OTP smart wallets remove the crypto barrier for farmers.

◆ Unique selling proposition

One protocol combines edge-AI quality gating, QR anti-cloning and gasless onboarding, with payment splits that reward quality.

CORE FEATURE

AI Quality Oracle + QR Anti-Cloning

KEY BENEFITS

Fair pay · Verified quality · ₹0 gas

TARGET USER

Farmers · Consumers · Retailers

Methodology & Tech Stack

From voice-listed crop to verified payout

◆ How it works, step by step

Voice + photo listing → AI quality score (below 40% is blocked) → on-chain batch record → QR scan check → escrow payout.

◆ Technologies used

TensorFlow.js MobileNet v2 · Solidity + Hardhat · Ethers.js v6 · ERC-4337 · Firebase · Express + Socket.io

◆ Implementation strategy

Modular monorepo with 4 smart contracts, 5 role-based dashboards and in-browser AI modules, plus a scripted demo for each feature.

TECH STACK

UI

Vite + Vanilla JS · Chart.js · EN / HI / TA / TE

Backend

Node.js · Express · Socket.io · REST API

Database

Firebase Firestore + Auth · JSON file store

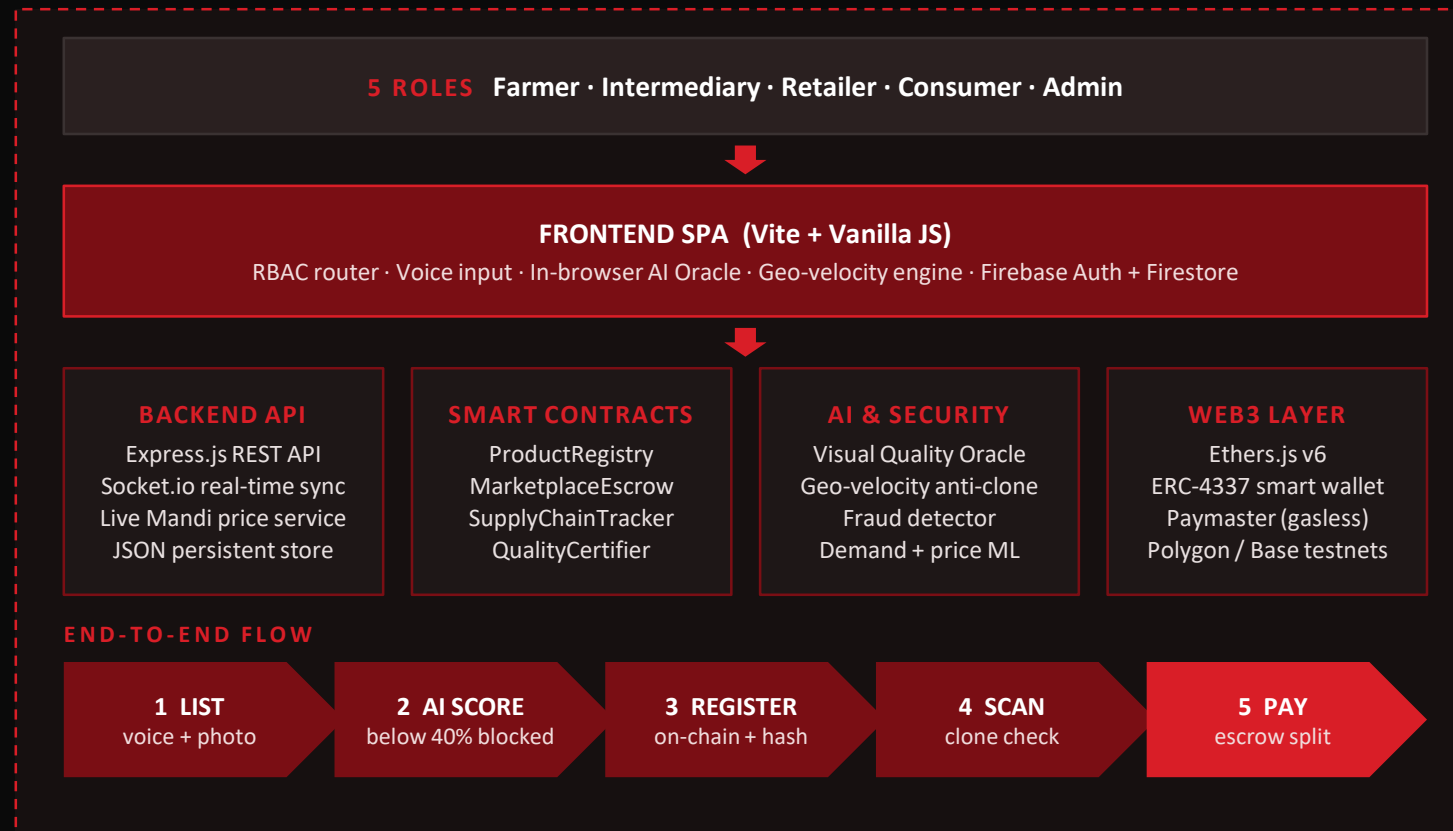
Core Logic (AI + Blockchain)

TF.js MobileNet v2 · Solidity · ERC-4337

SECTION 04

System Architecture

Trust is checked at every hop: AI, scan and escrow



KEEP IN MIND

- ◆ **Workflow:** List → AI score → Register → Trace → Pay
- ◆ **Data flow:** image hash → IPFS CID → on-chain record → QR → scan log → escrow split
- ◆ **Feasibility:** open-source stack, runs locally, testnet-ready

SECTION 05

Impact & Viability

Fairer pay for farmers, verifiable trust for buyers

EXPECTED IMPACT

- ◆ **Who benefits:** farmers, consumers, retailers and intermediaries (transparent margins)
- ◆ **Social & economic impact:** fairer farmer share, ₹0-gas onboarding, voice + regional-language access
- ◆ **Measurable outcomes:** farmer share 60% → 65% for 95%+ quality; sub-40% listings blocked; AI score + hash on every batch

USE CASE

A farmer lists 500 kg Basmati by voice and AI grades it. A consumer scans the QR and sees the verified journey. A copy scanned in another city too soon is flagged as a clone.

SCALABILITY

AI runs in the user's browser, so no server GPU is needed. Low-cost EVM networks (Polygon, Base) and modular roles and languages extend to new crops and regions.

FEASIBILITY

Working prototype with 4 smart contracts, 5 role dashboards and scripted demos, built on open-source tools. Testnet-ready.

THANK YOU

